VALORA FRONTEND ENGINEERING

Frontend Component & Screen Implementation Guide

Actionable Blueprint for Integrating New Backend Capabilities into the Group8Project UI

1. Purpose & Integration Scope

The upgraded backend serves expanded operational, analytical, and AI capabilities across 53 structured API endpoints. This guide outlines the specific **screens and components to add** to the existing frontend website, organized by **user role** and **location**.
Note: Per project requirements, the multi-estate switcher component is intentionally omitted. All screens described below integrate cleanly within the current single-community user workflow.

2. Master Component Reference Table

User Role	Component / Screen	Recommended Location	Primary Backend Endpoint
Resident	Incident Reporting Form	Resident Hub / Complaints Tab	POST /api/v1/incidents
Resident	Work Order Rating Dialog	Complaint Detail View (when done)	POST /api/v1/work-orders/{id}/rating
Resident	Self-Service Join Screen	Sign-up / Onboarding Flow	POST /api/v1/memberships/join
HOA Admin	Security Incidents Console	Admin Nav → /admin/incidents	GET/PATCH /api/v1/incidents
HOA Admin	AI Complaint Triage Widget	Complaint Review Detail Drawer	GET/POST /complaints/{id}/ai-suggestion
HOA Admin	Executive Health Brief	HOA Admin Main Dashboard	GET/POST /api/v1/analytics/health-brief
HOA Admin	Living Score Trend Chart	Dashboard Analytics Card	GET /analytics/living-score/history
HOA Admin	Estate Media & Profile Manager	Admin Nav → /admin/settings	GET/PATCH /api/v1/estates/profile
HOA Admin	Residency Approvals Console	Admin Nav → /admin/approvals	GET/POST /api/v1/residency-requests
Service Provider	Provider Assigned Work Queue	Provider Portal → /provider/work	GET /api/v1/work-orders/mine
Public / Visitor	Public Showcase & Gallery	Landing Page → /estates/:id	GET /api/v1/public/estates
Public / Visitor	Public Inquiry / Contact Form	Footer / Estate Detail Page	POST /api/v1/contact

3. Detailed Specifications by Role

• RESIDENT ROLE (Resident Experience)

1. Security Incident Reporting Form:
• *Where to place:* Add a "Report Security Incident" button in the Resident Complaints / Community view.
• *Fields:* Incident Type dropdown (Suspicious Activity, Noise, Theft, Property Damage), Severity (Low, Medium, High, Critical), Location, Date/Time, and narrative notes.
• *Endpoint:* POST /api/v1/incidents.
2. Work Order Post-Completion Star Rating Modal:
• *Where to place:* Appears automatically in the ticket detail view once a work order is marked COMPLETED.
• *Fields:* 1 to 5 star rating, feedback comment. Feeds the 15% satisfaction score in the Living Score engine.
• *Endpoint:* POST /api/v1/work-orders/{id}/rating.
3. Self-Service Join Community Screen:
• *Where to place:* In the resident sign-up flow or account profile (/join).
• *Fields:* 6-character estate join code input.
• *Endpoint:* POST /api/v1/memberships/join.

• HOA ADMINISTRATOR ROLE (Estate Operations)

4. Security Incidents Management Console:
• *Where to place:* Add "Security Incidents" to the Admin navigation sidebar (/admin/incidents).
• *Features:* Incident table with filters (Reported, Under Review, Resolved, Dismissed), severity tags, and a drawer to log resolution notes.
• *Endpoints:* GET /api/v1/incidents, PATCH /api/v1/incidents/{id}.
5. AI Complaint Triage Drawer Widget:
• *Where to place:* Side-drawer in the Admin Complaint Review screen.
• *Features:* Shows AI suggested trade (Plumbing, Electrical, etc.), urgency, and matching service provider. Actions: "Accept", "Override", "Dismiss".
• *Endpoints:* GET /api/v1/complaints/{id}/ai-suggestion, POST /api/v1/complaints/{id}/ai-suggestion/decision.
6. Executive Health Brief & Feedback Widget:
• *Where to place:* Top section of the HOA Admin Overview Dashboard.
• *Features:* Displays AI-generated cards: Headline, Key Drivers, and Recommended Actions. Includes Thumbs Up / Down feedback buttons.
• *Endpoints:* GET /api/v1/analytics/health-brief, POST /analytics/health-brief/{id}/feedback.
7. Living Score 90-Day Trend History Chart:
• *Where to place:* Directly beneath the Living Score card on the Admin Dashboard.
• *Features:* Line or area chart displaying score movement across the 5 pillars over the last 90 days.
• *Endpoint:* GET /api/v1/analytics/living-score/history.
8. Estate Media & Profile Manager:
• *Where to place:* Admin Settings menu (/admin/estate-settings).
• *Features:* Edit public description, community rules, and amenities. Manage photo gallery (upload, delete, set featured photo).
• *Endpoints:* PATCH /api/v1/estates/profile, POST /api/v1/estates/photos, DELETE /api/v1/estates/photos/{id}.
9. Residency Approvals Console:
• *Where to place:* Admin Navigation → Approvals (/admin/approvals).
• *Features:* Table of pending registration requests with one-click Approve / Reject buttons.
• *Endpoints:* GET /api/v1/residency-requests, POST /api/v1/residency-requests/{id}/approve.

• SERVICE PROVIDER & PUBLIC / VISITOR ROLES

10. Service Provider Work Queue (Provider Role):
• *Where to place:* Provider main screen (/provider/work).
• *Features:* List of assigned work orders, status buttons (Start Job → Complete Job), and resolution note entry.
• *Endpoints:* GET /api/v1/work-orders/mine, PATCH /api/v1/work-orders/{id}.
11. Public Estate Showcase (Visitor Role):
• *Where to place:* Public website landing page (/estates and /estates/:id).
• *Features:* Public cards for published estates featuring verified Living Score badge, photo gallery, description, and contact form.
• *Endpoints:* GET /api/v1/public/estates, POST /api/v1/contact.

4. Frontend Implementation Tips & Best Practices

• **API Error Handling:** All new endpoints return standardized error envelopes ({ error: { code, message, request_id } }). Display user-friendly messages and log request_id for troubleshooting.
• **Form Validation:** Match backend constraints (incident descriptions 10-1000 chars, join codes exactly 6 alphanumeric chars, ratings integers 1-5).
• **Optimistic State Updates:** When an admin accepts an AI triage suggestion or updates a work order status, update the UI immediately to maintain a snappy, responsive feel.